SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: ARTIFICIAL INTELLIGENCE

COURSE CODE: CSA17

CONSTRAINT-BASED PROBLEM SOLVING

OUT PUT PHOTOS:

```
==============================================================
1. HOSPITAL DOCTOR SHIFT SCHEDULING (BACKTRACKING SEARCH)
==============================================================

Valid Schedule Found:

D1 -> Morning
D2 -> Afternoon
D3 -> Night

Constraint Verification:
✓ D1 is not assigned Night Shift
✓ D2 works before D3
✓ D3 is not assigned Morning Shift
✓ One Doctor per Shift
✓ All Doctors Assigned

Final Schedule:
[D1=Morning, D2=Afternoon, D3=Night]


==============================================================
2. ROBOT NAVIGATION (BREADTH FIRST SEARCH)
==============================================================

Grid Size : 5 x 5
Start     : (0,0)
Goal      : (4,4)

Shortest Path:
(0,0) -> (0,1) -> (0,2) -> (0,3) -> (0,4)
      -> (1,4) -> (2,4) -> (3,4) -> (4,4)

Path Cost = 8

Goal Reached Successfully


==============================================================
```

```
==============================================================
3. AUTONOMOUS RESCUE ROBOT (UNIFORM COST SEARCH)
==============================================================

Environment Type : Partially Observable

Selected Strategy:
Uniform Cost Search (UCS)

Rescue Path Found:

S -> 0 -> 0 -> 0 -> 0 -> 0 -> 0 -> 0 -> G

Node Expansion:
S (cost=0)
0 (cost=1)
0 (cost=2)
0 (cost=3)
0 (cost=4)
0 (cost=5)
0 (cost=6)
0 (cost=7)
G (cost=8)
```